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Probability Theory

Indtroduction

- The theory of probability plays a crucial role in making inferences.
- Probability measures how likely something is to occur.
- Probability is the ratio of the number of favourable cases to the total number of cases, assuming that all of the various cases are equally possible.

Terminology

Experiment	• A process to yield an outcome. • Example: Rolling a dice.
Sample space	• A set of all possible outcomes from an experiment. • For a dice, that would be values 1 to 6.
Element	• An item in the sample space.
Event	• An outcome or combination outcomes from an experiment. • Example: Roll a 4 on the dice, the event = the 4.

How to Compute Probability

- We assign a probability weight $P(x)$ to each element of the sample space.
- Weight represents what we believe to be relative likelihood of that outcome.
- There are two rules in assigning weight:
  - i) The weight must be non-negative numbers.
  - ii) The sum of the weights of all the elements in a sample space must be 1.

- Let E be an event.
- The probability of *E*, *P(E)* is

$$P(E) = \sum_{x \in E} P(x)$$

- We read this as *"P(E) equals the sum, over all x such that x is in E, of P(x)"*

Probability Axioms

Let S be a sample space. A probability function P from the set of all events in S to the set of real numbers satisfies the following three axioms:

For all events A and B in S,
1. $0 \le P(A) \le 1$
2. $P(\emptyset) = 0$ and $P(S) = 1$
3. If A and B are disjoint, the $P(A \cup B) = P(A) + P(B)$

A function P that satisfies these axioms is called a probability distribution or a probability measure.

Complementary Probabilities

- The complement of an event A in a sample space S is the set of all outcomes in S except those in A.

$$P(A') = 1 - P(A)$$

- **Example:**
  Let the probability for getting a prize in a lucky draw is 0.075. Thus, the probability of *not* getting a prize in a lucky draw is 1 - 0.075 = 0.925.

a) Let A and B are events in a sample space S and suppose that $P(A') = 0.25$, $P(B) = 0.7$, and $P(A' \cup B') = 0.45$. What is $P(A \cap B)$? (4 marks)

b) Four people have been exposed to Corona virus. Once exposed, a person has a 50-50 chance of actually becoming ill.

i. What is the probability that exactly one of the people becomes ill? (3 marks)

ii. What is the probability that none of the four people becomes ill? (2 marks)

c) A box contains 20 green balls and 25 red balls. Two balls are chosen at random, one after the other, without replacement?

i. What is the probability that both balls are red? (2 marks)

ii. What is the probability that the second ball is red but the first ball is not? (2 marks)

d) A professor uses two proofreaders Ali and Hana to check a certain manuscript. Ali misses 9% of typographical errors and Hana misses 8%. Assume that the proofreaders work independently.

i. What is the probability that a randomly chosen typographical error will be missed by both proofreaders? (2 marks)

ii. If the manuscript contains 1000 typographical errors, what number can be expected to be missed? (2 marks)

Example 22

Mr. Basyir needs two students to help him with a science demonstration for his class of 18 girls and 12 boys. He randomly choose one student who comes to the front of the room. He then chooses a second student from those still seated. What is the probability that both students chosen are girls?

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